

Computational Wind Engineering

Bluff body aerodynamics

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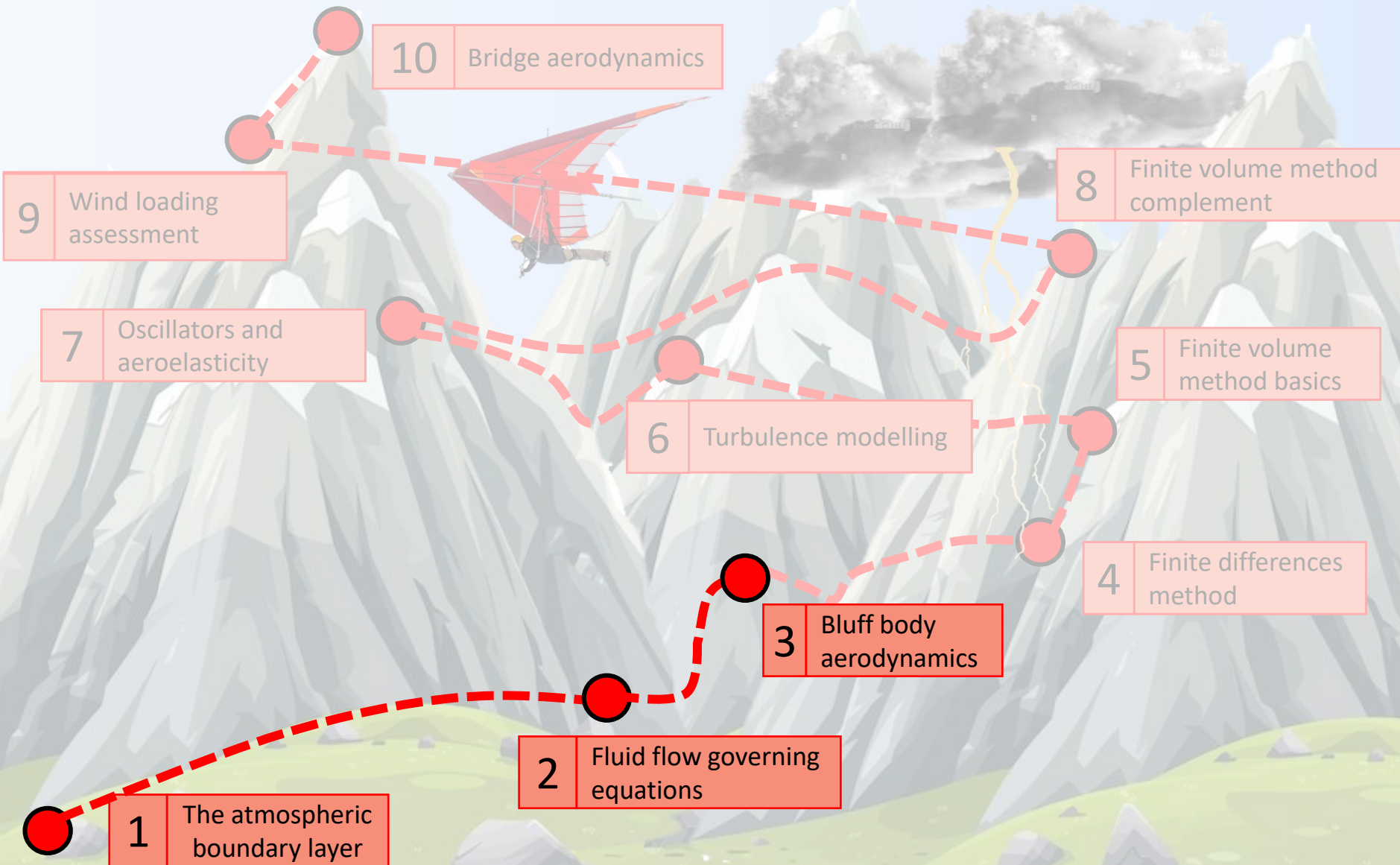
- The boundary layer and the law of the wall
- Non-dimensional Navier-Stokes
- Bernoulli's theorem
- Flow around bluff bodies
- Vortex shedding

Map of the course

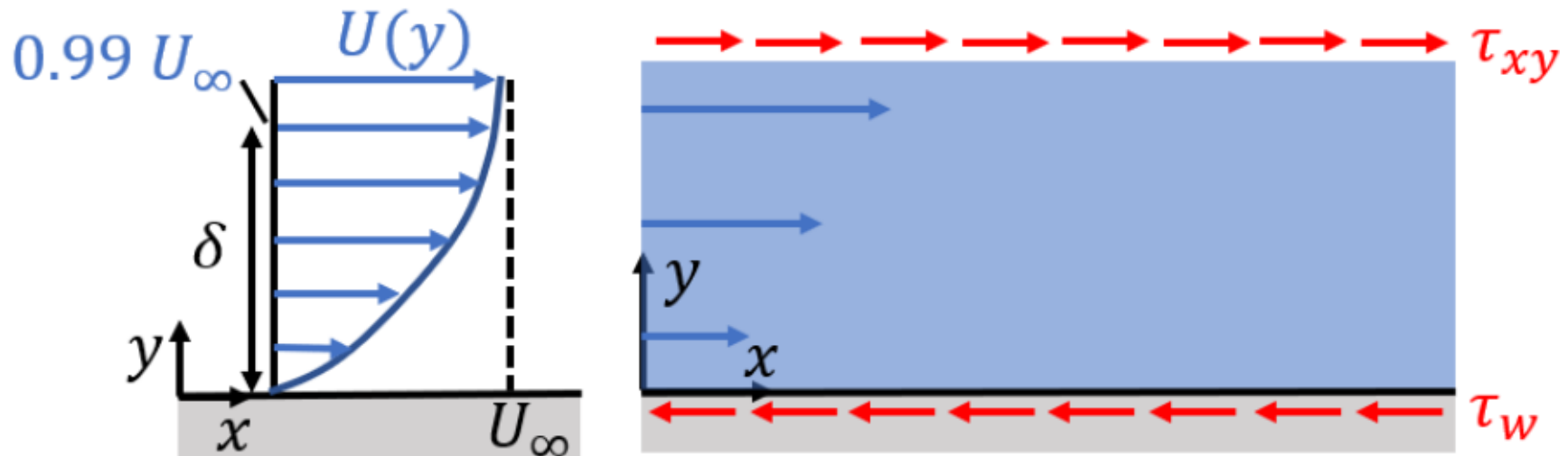
Wind Engineering

Fluid Dynamics

Numerical methods



We start by considering the flow established in the immediate proximity of the solid boundaries of bodies immersed in a fluid. In particular, we assume to remove all turbulent fluctuations from the flow field, and **describe only its time-average part**.



Neglecting pressure gradients, we have

$$\tau_w = \tau_{xy} \quad \longrightarrow \quad \tau_w = (\mu + \mu_t) \frac{dU}{dy}.$$

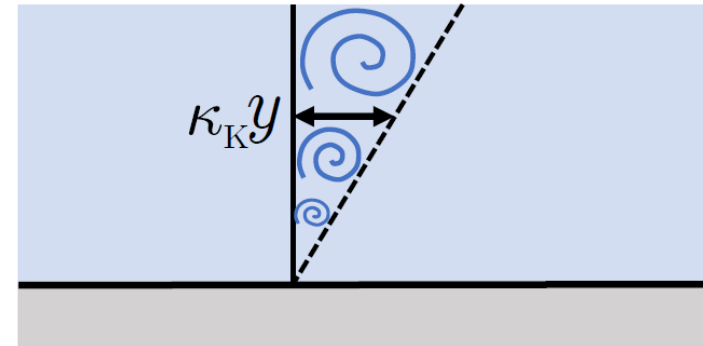
We firstly assume $\mu_t \gg \mu$ and according to the **mixing-length model**

$$\mu_t = C_\mu \rho l_t u_t.$$

Prandtl and von Kármán postulated

$$l_t = \kappa_K y$$

$\kappa_K \approx 0.41$ is the von Kármán constant



Based on dimensional ground

$$u_t = l_t \left| \frac{dU}{dy} \right| = \kappa_K y \left| \frac{dU}{dy} \right|$$

and substituting

$$\tau_w = \mu_t \frac{dU}{dy} = \rho \kappa_K^2 y^2 \left| \frac{dU}{dy} \right| \frac{dU}{dy}.$$

Disregarding signs, we extract the square root

$$\sqrt{\tau_w} = \sqrt{\rho} \kappa_K y \frac{dU}{dy},$$

arriving to

$$\frac{dU}{dy} = \sqrt{\frac{\tau_w}{\rho}} \frac{1}{\kappa_K y} = u_\tau \frac{1}{\kappa_K y}, \quad \Rightarrow \quad U(y) = \frac{u_\tau}{\kappa_K} \int \frac{1}{y} dy = \frac{u_\tau}{\kappa_K} \ln(y) + C^*,$$

where $u_\tau = \sqrt{\tau_w/\rho}$ is the **friction velocity**. Further, we define a length scale based on viscosity as

$$\mu / (\rho u_\tau).$$

and we rewrite the previous one as

$$u^+ = \frac{1}{\kappa_K} \ln(y) - \frac{1}{\kappa_K} \ln\left(\frac{\mu}{\rho u_\tau}\right) + C = \frac{1}{\kappa_K} \ln\left(\frac{\rho u_\tau y}{\mu}\right) + C = \frac{1}{\kappa_K} \ln(y^+) + C,$$

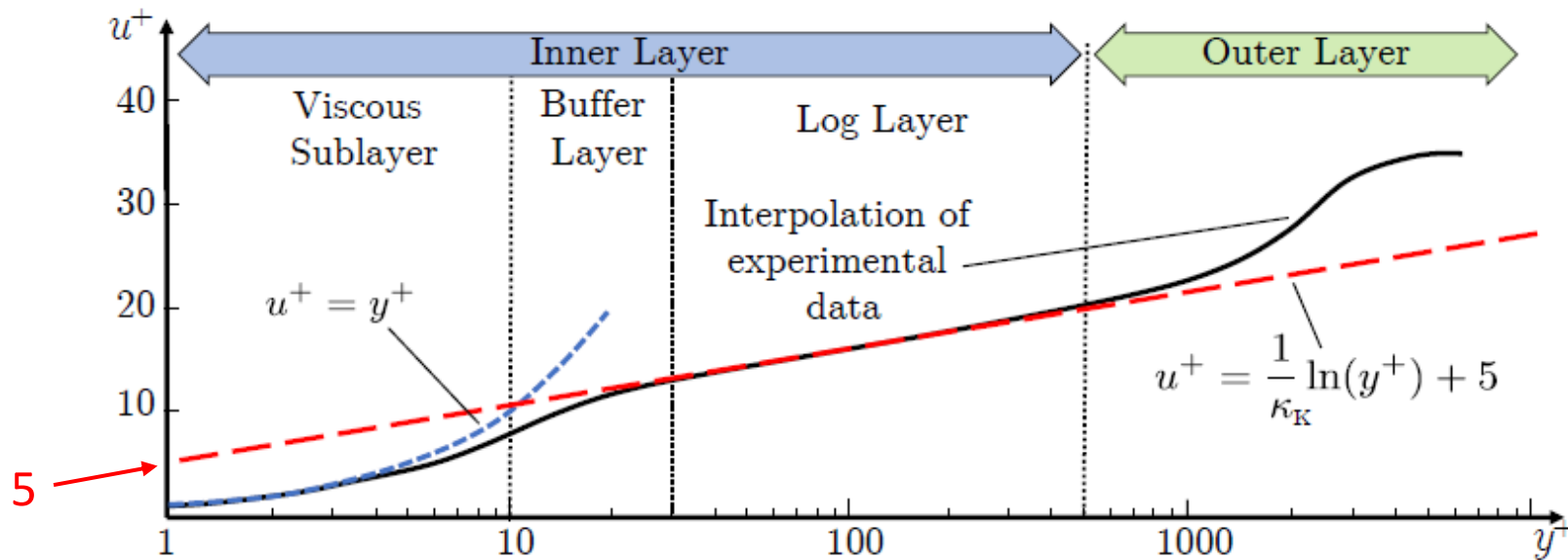
with

$$u^+ = \frac{U}{u_\tau}, \quad y^+ = \frac{\rho u_\tau y}{\mu},$$

While for $\mu \gg \mu_t$

$$\tau_w = \mu \frac{dU}{dy} \quad \Rightarrow \quad \rho u_\tau^2 = \mu \frac{dU}{dy} \quad \Rightarrow \quad 1 = \frac{d(U/u_\tau)}{d(y\rho u_\tau/\mu)} = \frac{du^+}{dy^+}$$

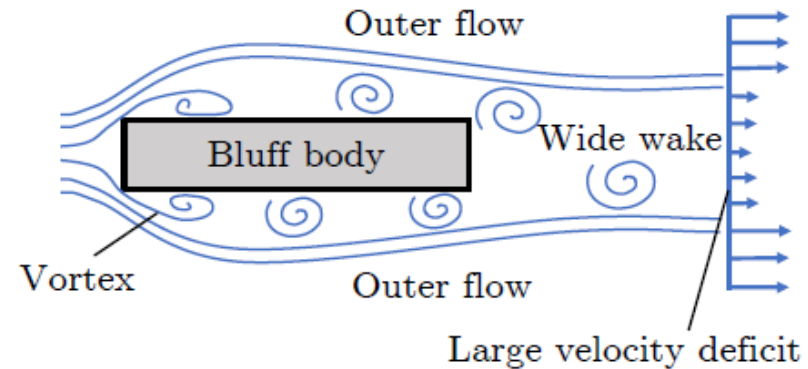
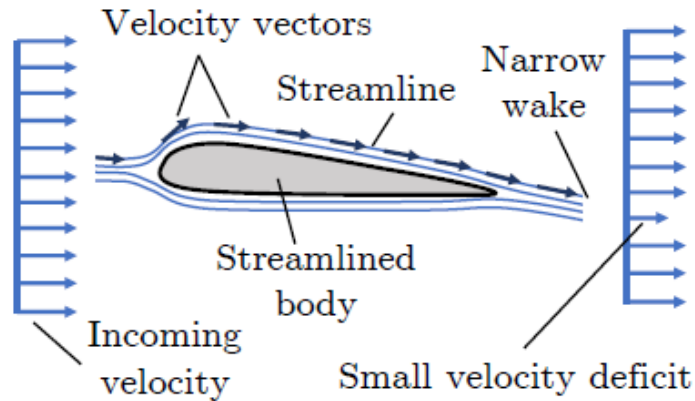
We thus arrive to the **law of the wall**



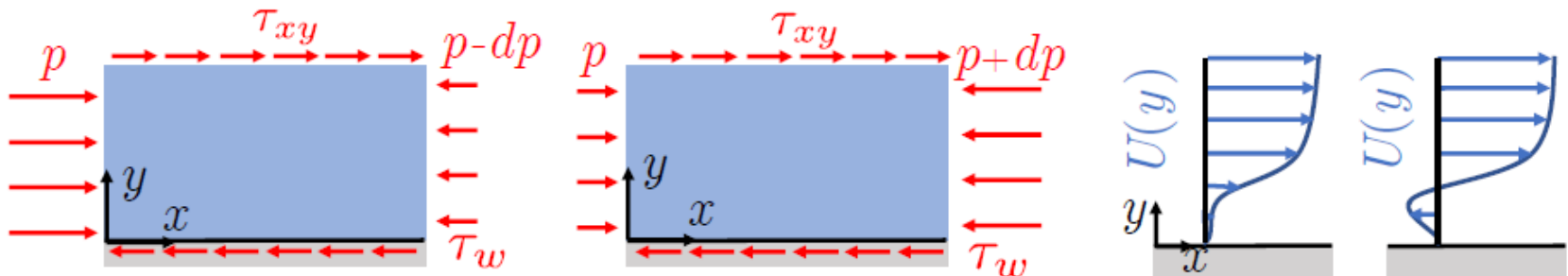
which is very well representative of the flow in the proximity of flat solid walls, provided pressure gradients are sufficiently small.

Streamlined vs Bluff I

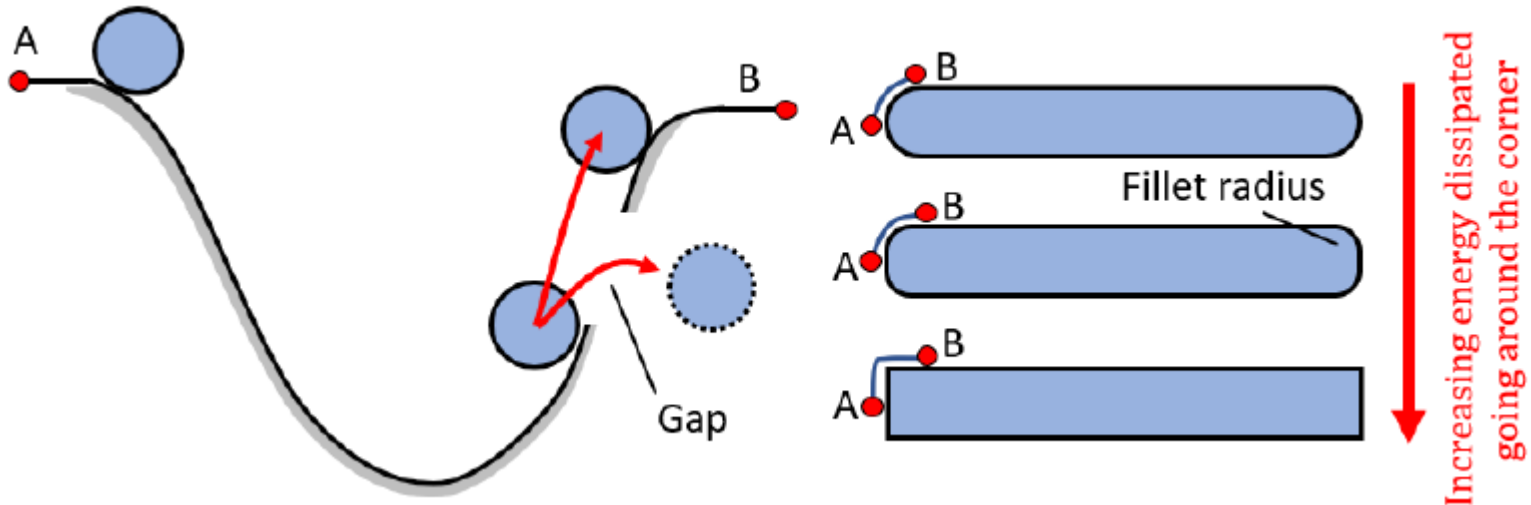
If the boundary layer stays attached to the body, we speak about streamlined bodies, bluff-bodies otherwise.



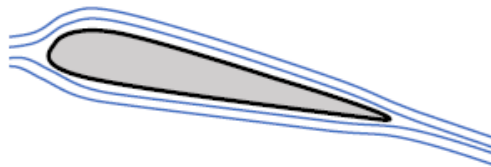
The boundary layer usually detaches due to adverse pressure gradients



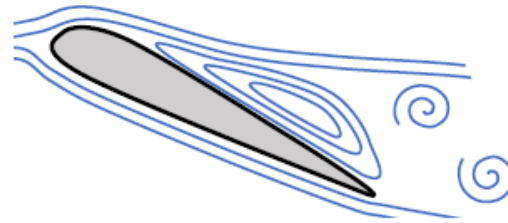
The boundary layer detachment is a runaway phenomenon



When the boundary layer detaches from a wing we say that it stalls



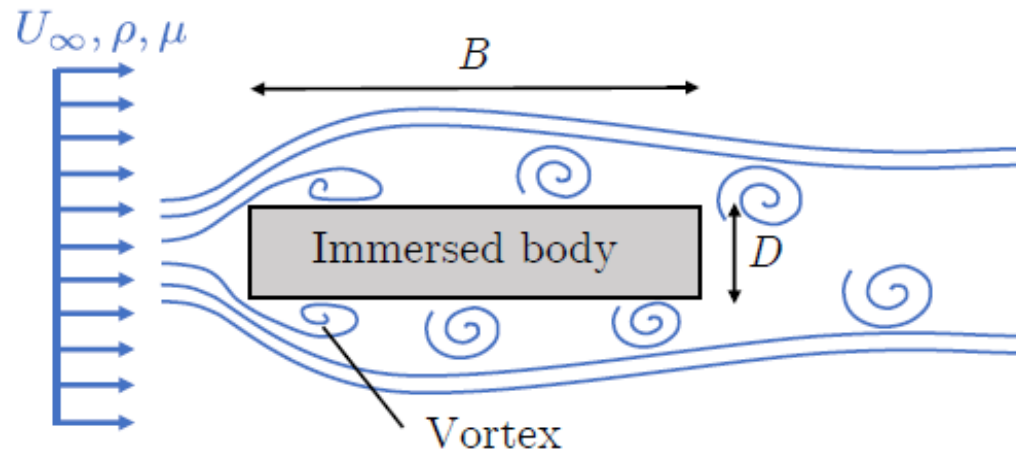
Correctly operating



Stalling

We now want to show that **the solution of Navier-Stokes equations depends only on one non-dimensional parameter**, the Reynolds number.

$$\begin{cases} \nabla \cdot \mathbf{u} = 0 \\ \frac{\partial \mathbf{u}}{\partial t} + \nabla(\mathbf{u})\mathbf{u} + \frac{\nabla p}{\rho} - \frac{\mu}{\rho} \nabla^2(\mathbf{u}) = 0. \end{cases}$$



We make the quantities appearing in the equations non-dimensional

$$\mathbf{u} = U_\infty \hat{\mathbf{u}}, \quad \mathbf{x} = B \hat{\mathbf{x}}, \quad t = \frac{B}{U} \hat{t}, \quad p = \rho U_\infty^2 \hat{p}.$$

Further, we make also the differential operators non-dimensional

$$\frac{\partial}{\partial t} = \frac{U_\infty}{B} \frac{\partial}{\partial \hat{t}}, \quad \nabla = \frac{1}{B} \hat{\nabla}, \quad \nabla^2 = \frac{1}{B^2} \hat{\nabla}^2.$$

We can thus arrive to

$$\begin{cases} \frac{1}{B} \hat{\nabla} \cdot U_\infty \hat{\mathbf{u}} = 0 \\ \frac{U_\infty}{B} \frac{\partial(U_\infty \hat{\mathbf{u}})}{\partial \hat{t}} + \frac{1}{B} \hat{\nabla} (U_\infty \hat{\mathbf{u}}) U_\infty \hat{\mathbf{u}} + \frac{1}{B} \frac{\hat{\nabla}[\rho U_\infty^2 \hat{p}]}{\rho} - \frac{\mu}{\rho} \frac{1}{B^2} \hat{\nabla}^2 (U_\infty \hat{\mathbf{u}}) = \mathbf{0}. \end{cases}$$

$$\begin{cases} \frac{U_\infty}{B} \hat{\nabla} \cdot \hat{\mathbf{u}} = 0 \\ \frac{U_\infty^2}{B} \frac{\partial(\hat{\mathbf{u}})}{\partial \hat{t}} + \frac{U_\infty^2}{B} \hat{\nabla}(\hat{\mathbf{u}}) \hat{\mathbf{u}} + \frac{\rho U_\infty^2}{\rho B} \hat{\nabla} \hat{p} - \frac{\mu U_\infty}{B^2 \rho} \hat{\nabla}^2(\hat{\mathbf{u}}) = \mathbf{0}. \end{cases}$$

$$\begin{cases} \hat{\nabla} \cdot \hat{\mathbf{u}} = 0 \\ \frac{\partial(\hat{\mathbf{u}})}{\partial \hat{t}} + \hat{\nabla}(\hat{\mathbf{u}}) \hat{\mathbf{u}} + \hat{\nabla} \hat{p} - \frac{1}{Re} \hat{\nabla}^2(\hat{\mathbf{u}}) = \mathbf{0}. \end{cases}$$

showing that the Reynolds number is the only parameter affecting the solution.

Another theorem of fundamental importance is **Bernoulli's Theorem**. We again from steady Euler equations, i.e. Navier-Stokes without time derivatives and viscous terms and, for the sake of simplicity, disregard source terms

$$\nabla(u)u + \frac{1}{\rho}\nabla p = 0$$

We project the momentum equations over the streamline tangent direction

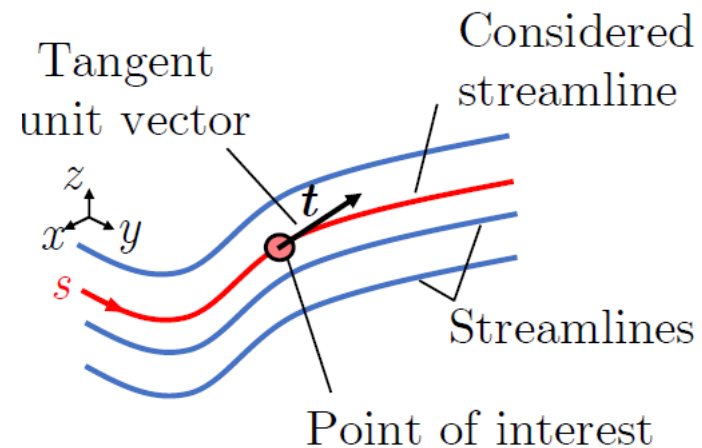
$$t^T \nabla(u)u + \frac{1}{\rho} t^T \nabla p = t^T 0.$$

$$\frac{u^T}{|u|} \nabla(u)u + \frac{1}{\rho} t^T \nabla p = t^T 0,$$

$$u^T \nabla(u)t + \frac{1}{\rho} t^T \nabla p = t^T 0.$$

$$u^T \left(\frac{du}{ds} \right) + \frac{1}{\rho} \frac{dp}{ds} = 0,$$

$$\frac{du}{ds}u + \frac{dv}{ds}v + \frac{dw}{ds}w + \frac{1}{\rho} \frac{dp}{ds} = 0.$$



For each contribution in the previous summation, we consider that

so that
$$\frac{d(u^2)}{ds} = 2u(s) \frac{du}{ds},$$

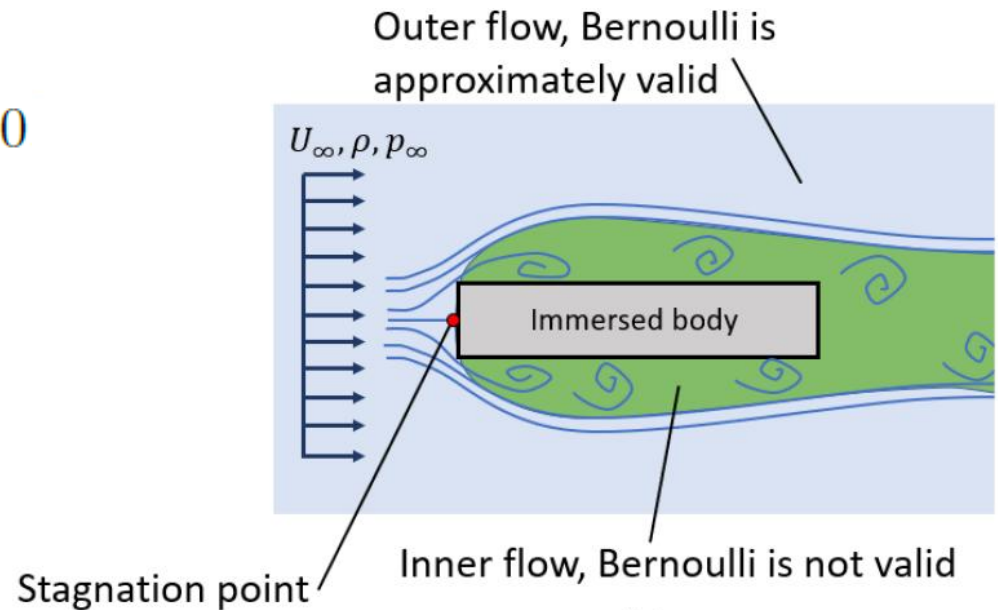
$$\frac{du}{ds}u + \frac{dv}{ds}v + \frac{dw}{ds}w + \frac{1}{\rho} \frac{dp}{ds} = 0. \quad \Rightarrow \quad \frac{d}{ds} \left(\frac{u^2 + v^2 + w^2}{2} \right) + \frac{1}{\rho} \frac{dp}{ds} = 0.$$

Considering that $u^2 + v^2 + w^2 = |\mathbf{u}|^2$ we can integrate over the streamline

$$\int_{s_1}^{s_2} \left[\frac{d}{ds} \left(\frac{|\mathbf{u}|^2}{2} \right) + \frac{1}{\rho} \frac{dp}{ds} \right] ds = 0$$

$$\left[\frac{|\mathbf{u}|^2}{2} + \frac{p}{\rho} \right]_{s_1}^{s_2} = 0,$$

$$\frac{1}{2} |\mathbf{u}|^2 + \frac{p}{\rho} = \text{cons.}$$



Let us now consider the streamline which arrives to the stagnation point

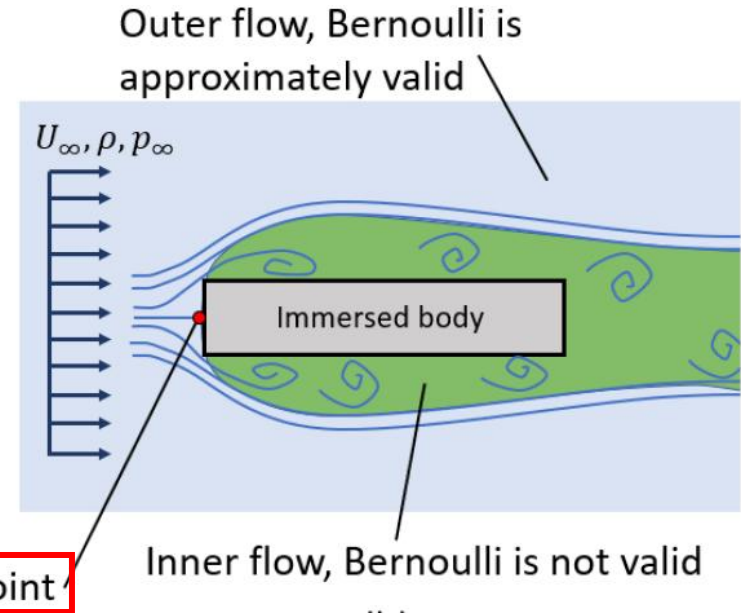
$$\frac{1}{2}|U_\infty|^2 + \frac{p_\infty}{\rho} = \frac{p_{st}}{\rho},$$

$$p_{st} - p_\infty = \frac{1}{2}\rho|U_\infty|^2.$$

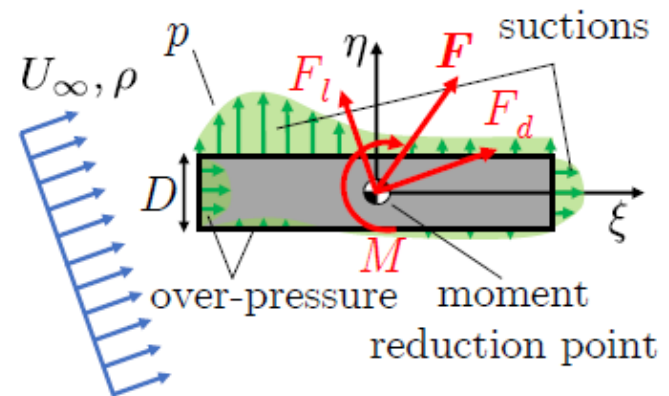
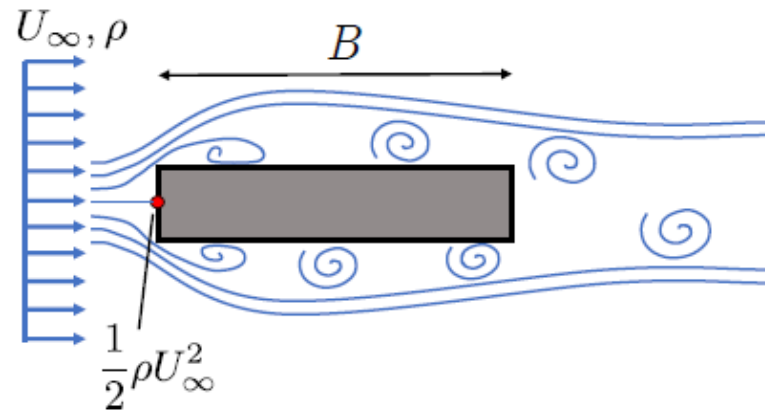
Taking as reference $p_\infty = 0$, Stagnation point, we define the dynamic pressure as the pressure at the stagnation point

$$q = \frac{1}{2}\rho|U_\infty|^2.$$

The dynamic pressure based on a reference velocity is usually taken as reference pressure. It can be seen that it is half the pressure scale previously used in making Navier-Stokes non-dimensional.



Let us first consider in a qualitative way the flow around a general bluff-body, with special attention to all the quantities which characterize the observed flow field.



We want to make pressure non-dimensional. To this purpose, we use the dynamic pressure, so that

$$c_p = p/q,$$

being c_p the pressure coefficient.

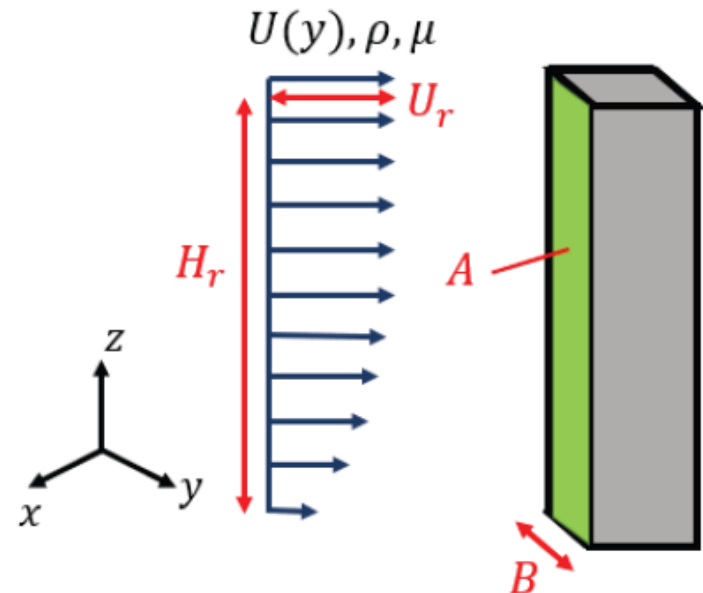
We can also make the measured **forces non-dimensional** as

$$C_d = \frac{F_d}{qB} = \frac{F_d}{1/2\rho U_\infty^2 B}, \quad C_l = \frac{F_l}{qB} = \frac{F_l}{1/2\rho U_\infty^2 B}, \quad C_m = \frac{M}{qB^2} = \frac{M}{1/2\rho U_\infty^2 B^2}.$$

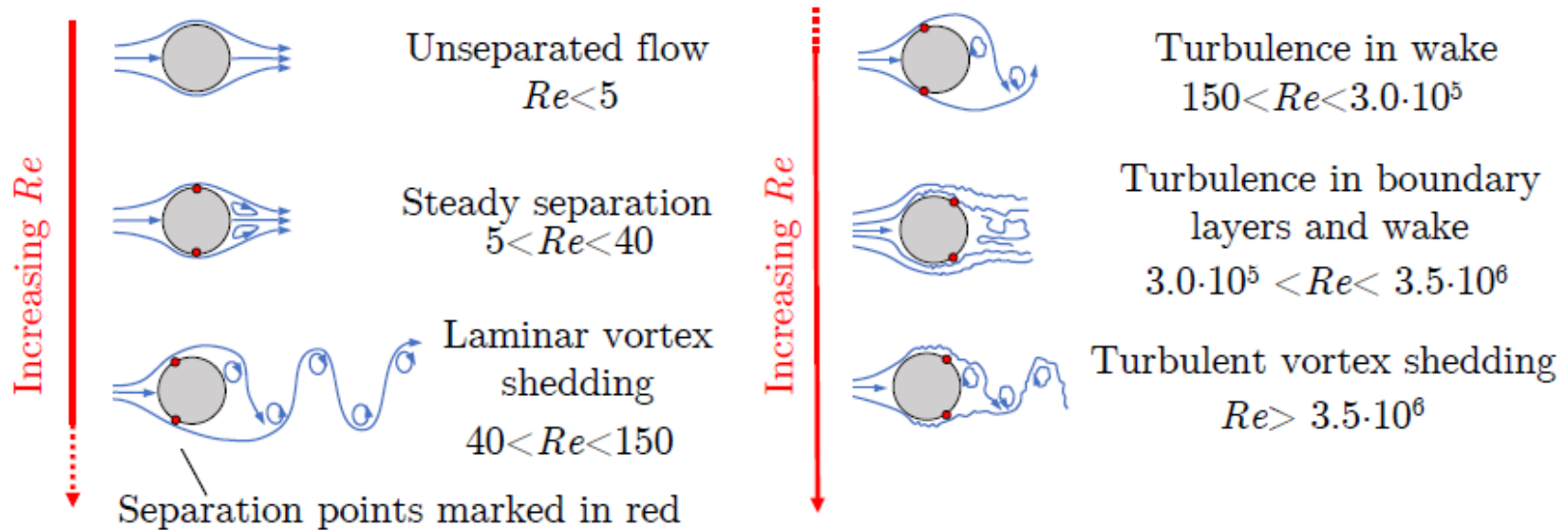
or adoption the section reference system

$$C_\xi = \frac{F_\xi}{qB}, \quad C_\eta = \frac{F_\eta}{qB}.$$

In a general case of a 3d flow, we proceed selecting a reference velocity, measured in the undisturbed flow upstream the body, and calculate coefficients in a similar way.

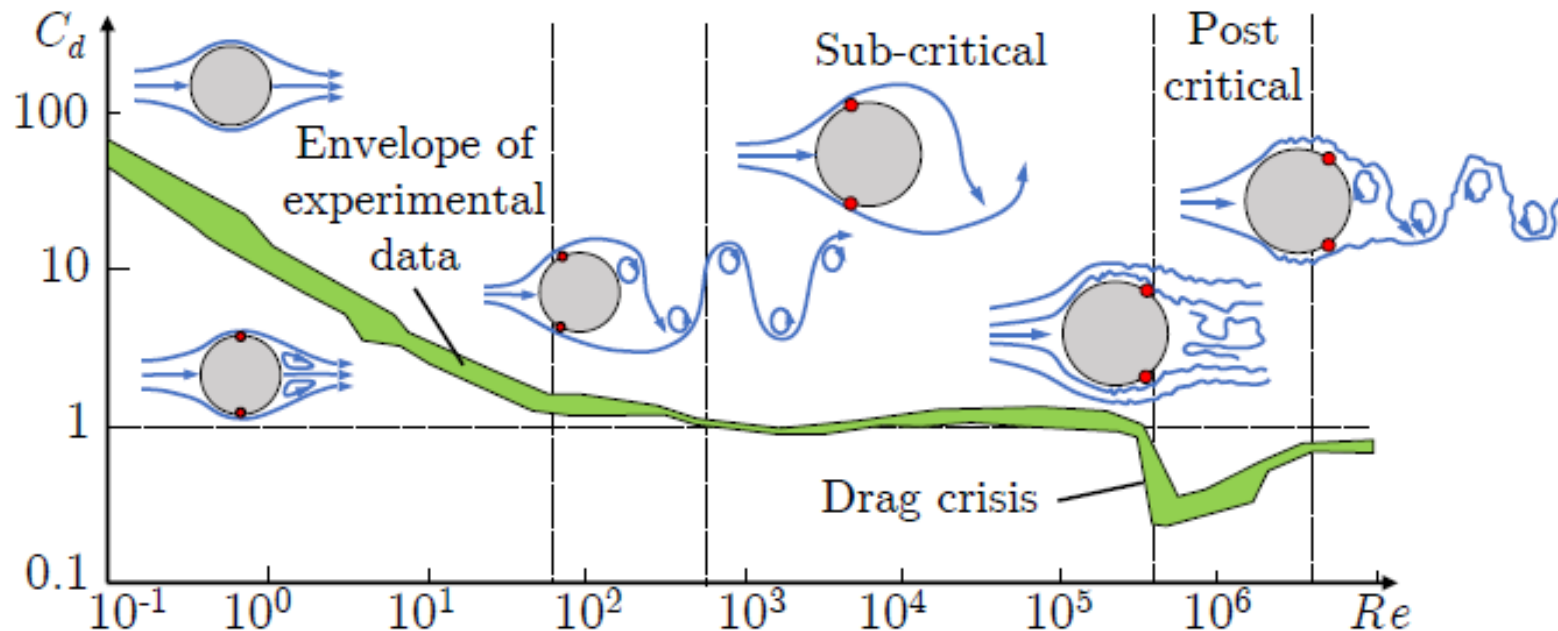


We already demonstrated that the only free parameter affecting the flow field is the Re number. The circular cylinder is emblematic to show its effect.

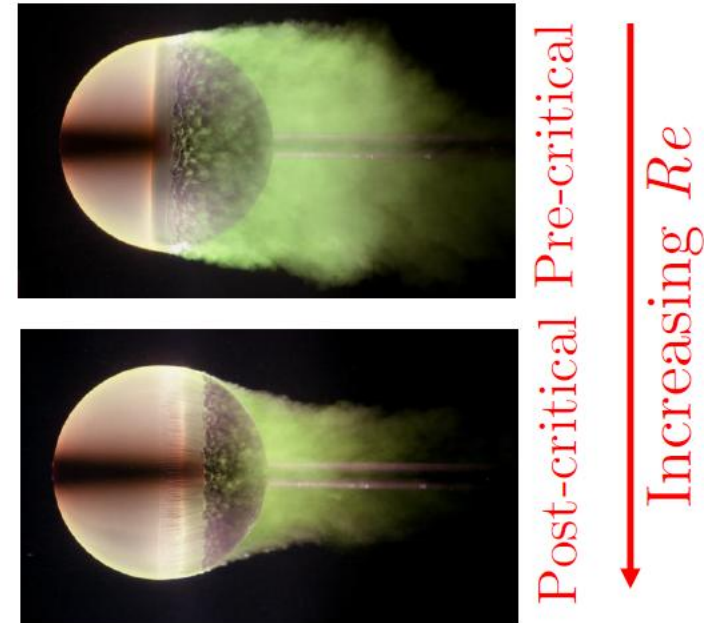
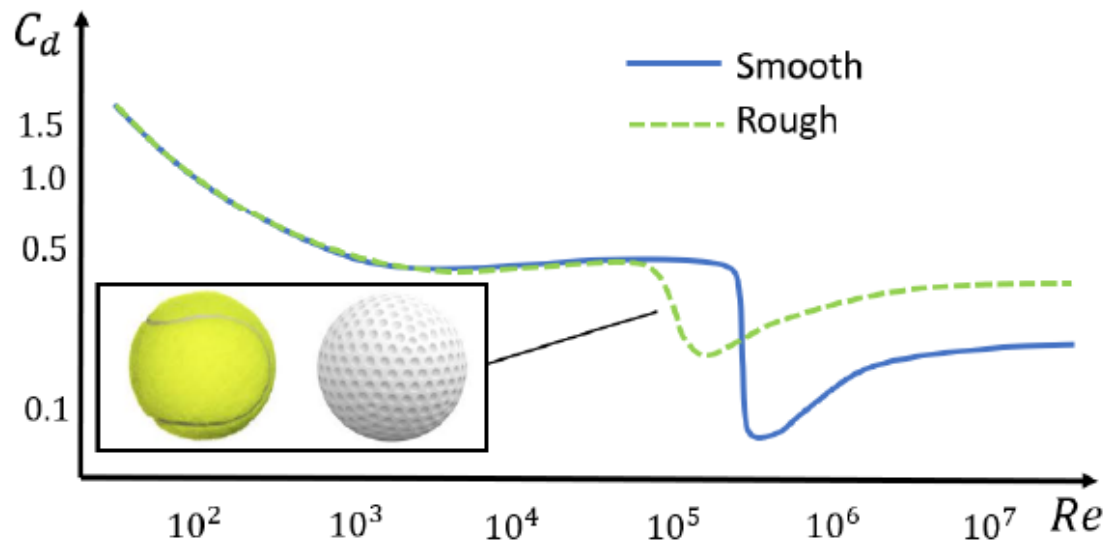


The Re number affects the position of the separation points and the wake development.

The modification of the flow field has direct effects on the forces recorded over the body.

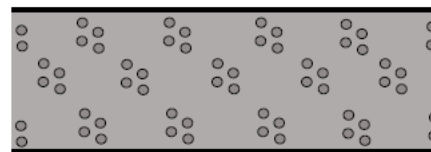


An effect similar to that produced by increasing Re can be obtained varying the immersed body **roughness**.



Photographs taken by Werl'e in 1980, collected in An Album of Fluid Motion (Van Dyke 1982), courtesy of ONERA.

Increasing roughness is commonly used in playing balls as well as bridge cables.



Let us now analyse some recurrent features around **sharp-edged bodies**, starting from geometries obtained from extrusion of a planar shape (cylinders of varying cross sections)

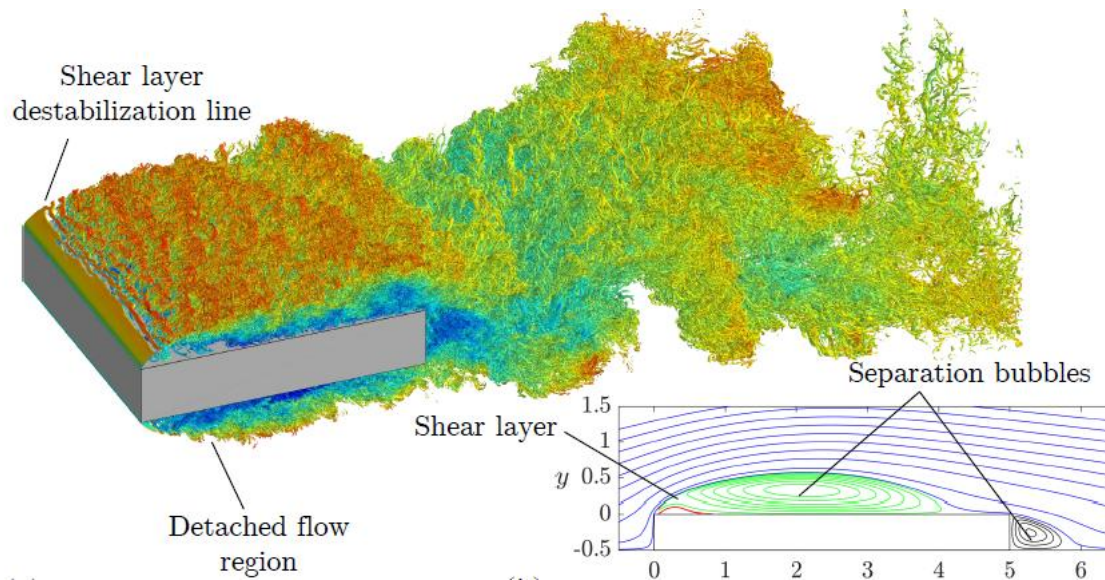
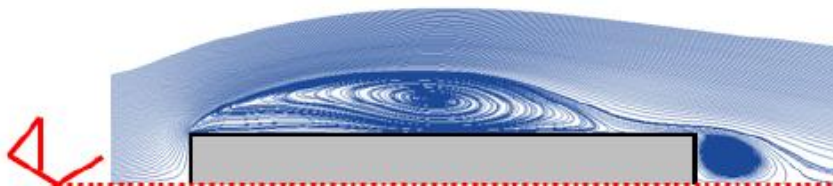


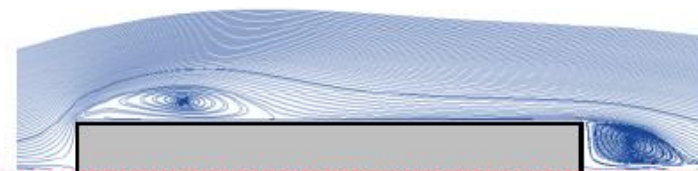
Image courtesy of A. Cimarelli.

Incoming smooth flow

Incoming turbulent flow $I_u = 13.6\%$

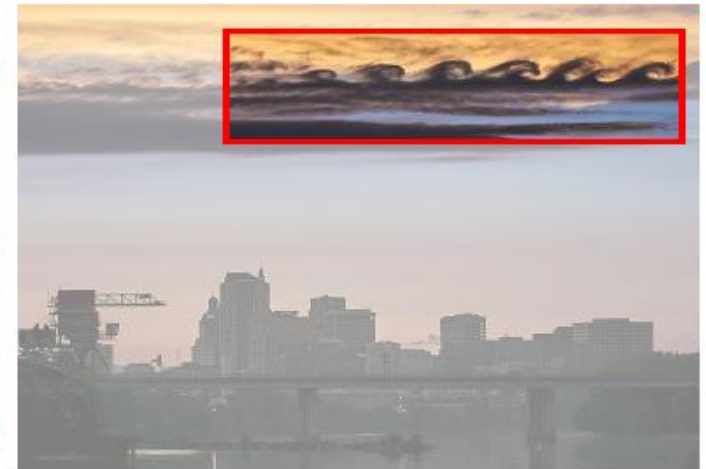
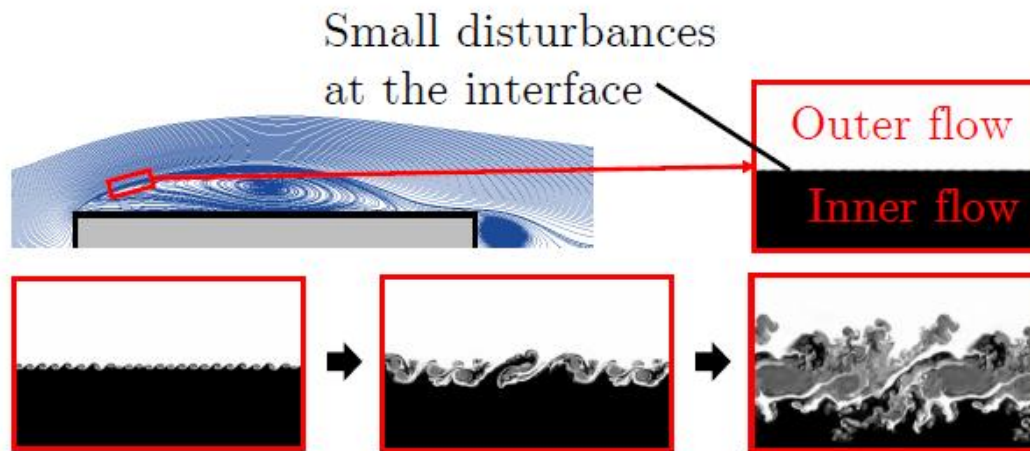


(a)



(b)

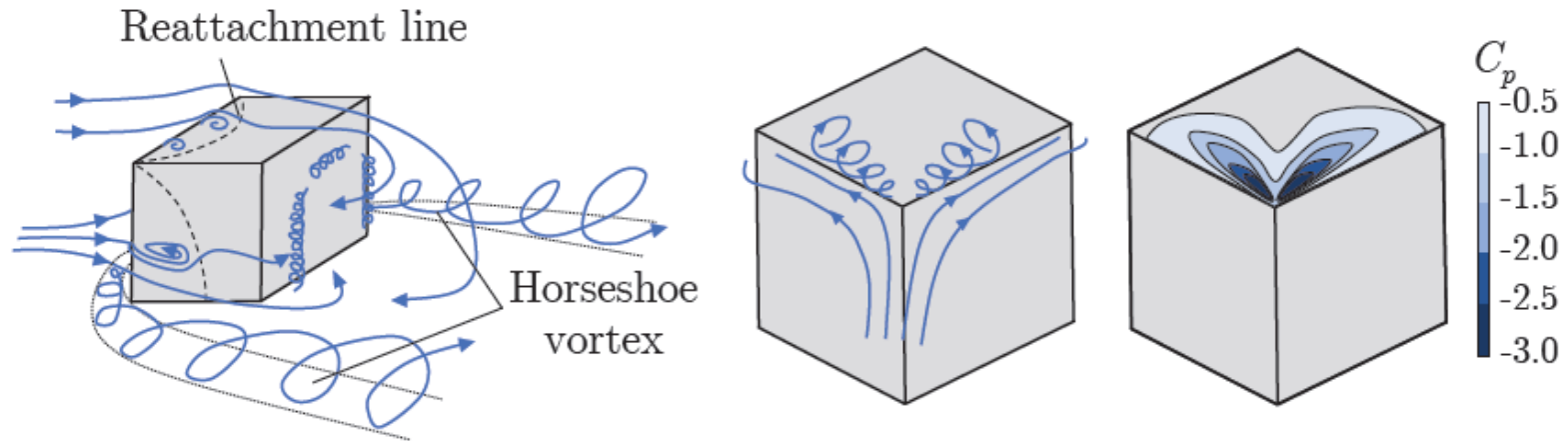
One of the most important mechanisms which lead to the destabilization of shear layers is the **Kelvin-Helmholtz instability**.



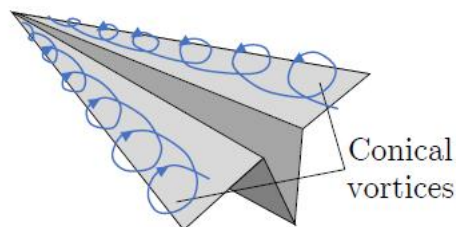
Images adapted from public domain material available on Wikimedia Commons.

Once some disturbances are present at the interface between the high speed and the low speed zones, depending on the wavelength, the velocity difference and the fluid viscosity, **these disturbances might be damped out or amplified**. In the latter case, turbulence will be finally produced.

Let us now consider some simple three-dimensional flows



Conical vortices can produce very strong depression, and they are particularly common when the flow impinges diagonally on edges



Right image released in the public domain available online at [Wikimedia Commons](#).

We already observed that **vortices are periodically shed from immersed bodies**. Let us now consider it in more detail.

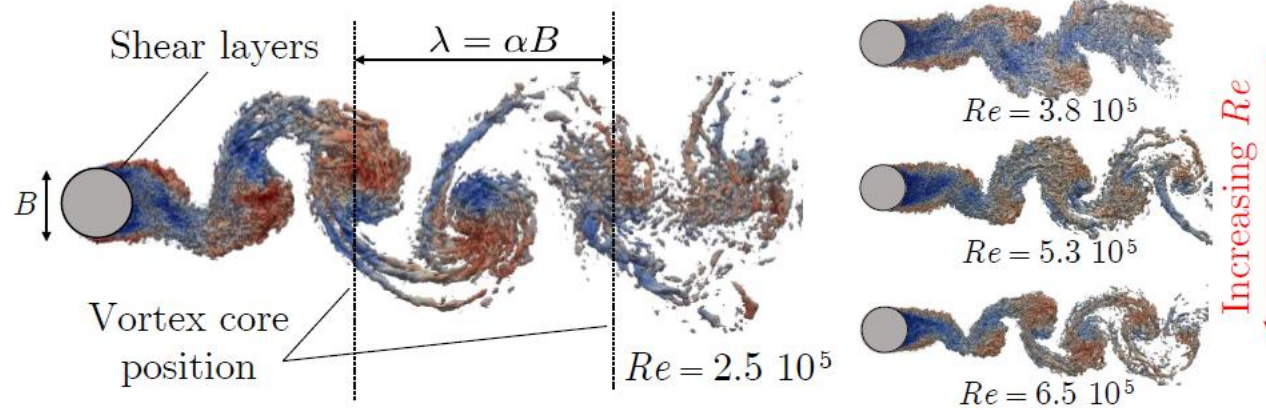


Image adapted from
O. Lehmkuhl et al.
2014 *Unsteady forces
on a circular cylinder
at critical Reynolds
numbers.*, courtesy of
the authors and with
the permission of AIP
Publishing.

Looking at the picture, we notice that

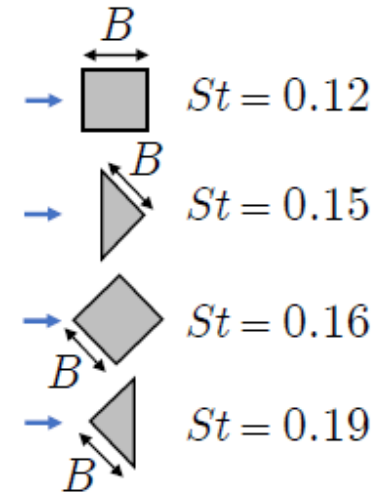
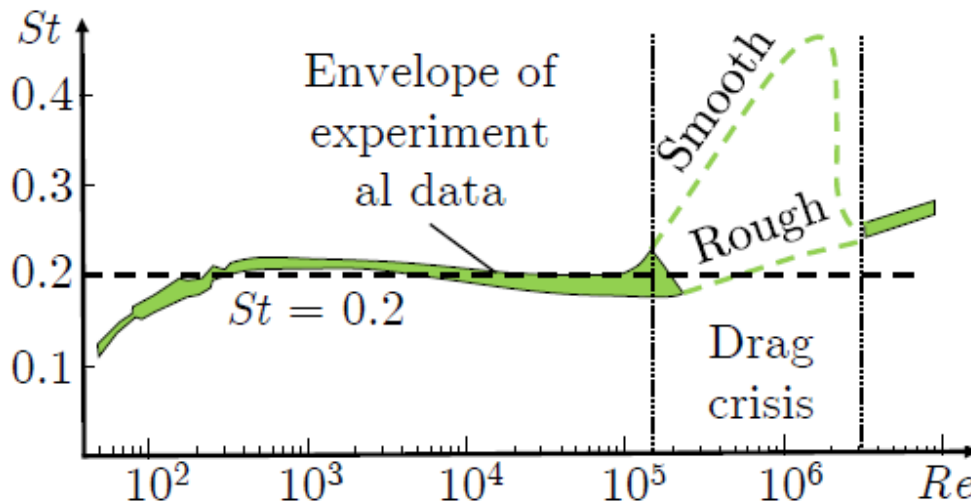
$$T_{vs} = \frac{\alpha B}{\beta U_{\infty}} = \frac{\alpha}{\beta} S \quad \longrightarrow \quad 1/T_{vs} = \beta/\alpha \frac{U_{\infty}}{B} = St \frac{U_{\infty}}{B},$$

Considering that period is the inverse of the frequency

$$f_{vs} = \frac{U_{\infty}}{B} St.$$

with St the Strouhal number
(non dimensional)

The Strouhal number depends only on the section and, eventually, Re but its values are often concentrated in a relatively small range.



A particularly critical condition is reached when vortex shedding synchronises with the structural vibration

$$f_s = \frac{U_{c,vs}}{B} St, \quad \longrightarrow \quad U_{c,vs} = \frac{f_s B}{St}.$$